# INTRODUCTION

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# ACTING UNDER UNCERTAINTY

- Uncertainty is a situation which involves imperfect and/or unknown information. However, uncertainty is an unintelligible expression without a straightforward description.
- Uncertainty can also arise because of incompleteness, incorrectness in agents understanding the properties of environment.
- So to represent uncertain knowledge, we are not sure about the predicates, we need uncertain reasoning or probabilistic reasoning.

## Causes of Uncertainty:

Following are some leading causes of uncertainty to occur in the real world:

- Information occurred from unreliable sources
- Experimental errors
- Equipment fault
- Human variation
- Precipitation

# PROBABILISTIC REASONING

- It is a technique used in AI to address uncertainty by modelling and reasoning with probabilistic information.
- Probabilistic reasoning is a way of knowledge representation where we apply the concept of probability to indicate the uncertainty in knowledge. In probabilistic reasoning, we combine probability theory with logic to handle the uncertainty.
- We use probability in probabilistic reasoning because it provides a way to handle the uncertainty that is the result of someone's laziness and ignorance.
- In the real world, there are lots of scenarios where the certainty of something is not confirmed. Such as: "It will rain today", "behavior of someone", or "some situations", a match between two teams or two players.
- These are probable Sentences for which we can assume that it will happen but not sure about it, so here we use probabilistic reasoning.

# PROBABILISTIC REASONING CONTD...

*Need of Probabilistic Reasoning:*

1. When there are unpredictable outcomes.
2. When specifications or possibilities of predicates become too large to handle.
3. When an unknown error occurs during an experiment.

In probabilistic reasoning, there are two ways to solve problems with uncertain knowledge:

1. Baye's rule
2. Bayesian statistics

# HANDLING UNCERTAIN KNOWLEDGE

Handling uncertain knowledge in AI refers to the ability of an AI system to make decisions or draw conclusions even when the information it has is incomplete, ambiguous, or unreliable.

In the real world, data is rarely perfect — Sensors might fail, user inputs might be vague (not clear), or the environment might change unexpectedly. AI systems need strategies to reason under uncertainty to still perform effectively.

# HANDLING UNCERTAIN KNOWLEDGE

## Methods for Handling Uncertainty

AI employs a variety of methods to manage and reason about uncertain knowledge, each with specific applications and strengths:

### 1. Probabilistic Reasoning

It involves representing knowledge using probability theory to manage uncertainty.

### 2. Hidden Markov Models

These are used to model time series data where the system being modeled is assumed to be a Markov process with hidden states. HMMs are widely used in speech recognition, bioinformatics, etc.

### 4. Fuzzy Logic

It is an approach to reasoning that deals with approximate values rather than fixed and exact values.

# UNCERTAINTY AND RATIONAL DECISIONS

The presence of uncertainty significantly impacts how an agent makes decisions. A logical agent typically has a clear goal and executes any plan that is guaranteed to achieve it.

In such cases, actions are chosen or rejected based on whether they accomplish the goal without considering alternative actions. However, when uncertainty is involved, this approach is no longer sufficient.

## Preferences and Utility Theory

- To make rational decisions under uncertainty, an agent must first establish preferences between different possible outcomes.
- An outcome represents a fully described state.
- Utility theory provides a way to represent and reason with these preferences. In this context, utility refers to the measure of how beneficial or desirable a particular outcome is to the agent.
- According to utility theory, every state has a certain degree of usefulness or utility, and a rational agent will always prefer states with higher utility.
- The utility of a state depends on the specific agent and their preferences.

# UNCERTAINTY AND RATIONAL DECISIONS

## Utility Function

It is a mathematical tool that assigns a value (or score) to different possible outcomes, reflecting how desirable they are to the AI's goals. It is like a scorecard that tells the AI what to aim for.

## How it works:

The AI evaluates each possible action it could take by estimating the utility of the resulting outcomes. Higher utility means a more desirable outcome.

- Ex: For a chess-playing AI, the utility function might assign:
- +100 for a win (the ultimate goal)
- 0 for a draw
- -100 for a loss

# UNCERTAINTY AND RATIONAL DECISIONS

## Decision Theory

Rational decision-making under uncertainty combines two key components:

**Decision Theory = Probability Theory + Utility Theory**

This approach differs from traditional logic, where a sentence becomes true or false as the world changes. Dealing with an uncertain, dynamic world requires the use of situations, actions, and events — concepts similar to those used for logical representations.

The core principle of decision theory is the maximum expected utility (MEU). According to this principle, an agent is rational if and only if it chooses the action that produces the highest expected utility averaged across all possible outcomes.

By combining probabilities with preferences, decision theory provides a powerful framework for making rational choices even in uncertain environments.

## Design for a Decision Theoretic Agent

A decision theoretic agent selects actions based on probabilities and expected utility rather than absolute certainty. It maintains a belief state representing the likelihood of different possible real-world states. As it gathers more evidence over time, it updates this belief state and makes probabilistic predictions about action outcomes — choosing the action with the highest expected utility.

# CONCLUSION

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